Define Problem Statements

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	3 Marks

1. Problem Context & Background

Modern law enforcement and forensic investigative agencies encounter vast amounts of heterogeneous digital evidence at crime scenes, including high-resolution crime scene photos, 911 dispatch audio recordings, witness statements, and multi-angle CCTV surveillance video. Currently, these streams remain siloed across disparate file folders and incompatible tools.

2. Core Problem Statements

ID	Core Problem Statement	Root Cause	Impact on Investigation
PS-01	**Fragmented Evidence Synchronization:** Visual evidence (photos), auditory calls (911/witness), and CCTV surveillance are analyzed in silos without a common temporal clock.	Absence of multimodal cross-referencing platforms capable of fusing time-stamped video with speech and 2D spatial evidence.	Critical timeline discrepancies, high investigator burnout, missed corroborating clues, and delayed judicial proceedings.
PS-02	**Loss of 3D Spatial Realism:** Flat 2D photographs fail to convey true physical distances, weapon trajectories, victim orientation, and room scale.	Reliance on static photography and manual tape-measure sketches without depth estimation or geometric 3D models.	Vulnerability to aggressive defense cross-examination in court regarding line-of-sight and physical plausibility.
PS-03	**Language Barriers in Witness Testimonies:** Eyewitnesses in multicultural urban zones speak regional/foreign languages that investigators cannot promptly translate.	Lack of automated, immediate forensic-grade multilingual speech-to-text and cross-lingual translation workflows.	Vital initial golden-hour investigative leads are lost while awaiting official accredited translators.
PS-04	**Inconsistent & Slow Forensic Documentation:** Generating comprehensive case summary reports requires hours of manual drafting prone to omission.	Manual collation of handwritten notes, detection printouts, and audio transcripts without cryptographic validation.	Chain-of-custody vulnerabilities and delays in submitting court-admissible dossiers to prosecutors.

3. Expected Objectives & Target Beneficiaries

• **Primary Objective:** Build an intelligent, multimodal AI forensic reconstruction platform that fuses visual, auditory, and surveillance evidence into an interactive 3D spatial canvas and synchronized timeline.
• **Beneficiaries:** Homicide & Cyber Forensic Squads, Police Investigators, District Attorneys, Court Jurors, and Digital Evidence Custodians.